JACOB LYNN

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Project Portfolio

Education

Stanford University Expected Graduation June 2029
B.S. Mechanical Engineering, GPA: 3.41
Minor in Computer Science
Relevant Coursework:
CS223A: Intro to Robotics, ME210: Intro to Mechatronics

Relevant Experience

ARMLab Researcher - Assistive Robotics and Manipulation Apr 2026 - Present

- Designed an electromagnetic linear actuator for a robotic muscle for tendon-driven robot hands with full position (PID) and force control; patent application filed (patent pending)
- Coded ROS2 environments for finger control and force simulation to determine electromagnet spacing using self-collected data
- Designed a sensor that calculates the position of the actuator with 0.1mm precision by measuring the changing inductance as permanent magnets move through it
- Prototype weighs 70g and delivers 5N continuously over a 15mm stroke; a 7:1 force-to-weight ratio
- Designed for a 16 actuator array (1 per finger joint) to control a fully dexterous robot hand
- Worked full time (40hr/week) for 10 weeks during the summer and 8hr/week during the school year

Stanford Student Space Institute - Europa Team Co-lead Sep 2025 - Present

- Managed a team of ∼10 people to design a low-cost (<$5K) open source benthic (ocean floor) lander rated for 1000m depths that collects sediment, DNA, water, and video
- Led the mechanical design, including a custom soil scoop, Niskin tube, and a pressure-powered DNA collector
- Ran 4 successful field tests in SF Bay and Monterey Bay (100m+) with MBARI (Monterey Bay Aquarium Research Institute); collected sediment samples, video, and GPS tracking data
- Prototyping for phase 2 to reach depths of +10,000m to reach the floor of the Mariana Trench by Fall 2027

Stanford Student Space Institute - IREC Team, Payload Subteam Sep 2025 - Present

- Designed an L3 rocket payload that synthesizes crystals in microgravity for pharmaceutical testing to compete in the International Rocket Engineering Competition (IREC)
- Engineered a rigid, compact payload and vibration-resistant interior structure that fit within a 2U CubeSat volume
- Designed two prototypes with a multi-tiered interior system for easy repair and modification

Projects

Modular Robot Dog Link to project

- Designed and tested a modular QDD actuator system with swappable gearboxes (including a 5:1 planetary gearbox, an 18:1 cycloidal gearbox, and a 22.5:1 harmonic gearbox)
- Built a full-sized 13.5lb 12DoF robot dog standing ∼ 6in off the ground; currently developing the electronics
- Created a modular system of actuators to act as a framework for limbed robots

Reconfigurable Robot Builder Website

- Coded a website where users can design any robot and export it as a STEP file with fully working motors and gearboxes
- Uses the modular gearbox system I designed in the Modular Robot Dog project
- Designed 7DoF robot arms, a mimic of JPL's RoboSimian, robot dogs, and hexapods using the website

Skills

Engineering	CAD, CAM (Fusion 360, Onshape)
Fabrication	CNC machining, 3D printing, laser cutting, carbon fiber/composite epoxy layups
Programming	Python, JavaScript, Java, C, C++, ROS2, Linux
Control/Embedded Systems	CAN, microcontrollers (Teensy 4.1, Arduino Uno, ESP32), PID, KiCad